

ProductionManagementAI — Identity Schema — Database Design Document (テーブル定義書)

000_DB — supports auth/RBAC foundation (`work-items/WI-001/brief.md` INFRA-002), implements 0002_ADR (`docs/en/architecture/0002/0002_ADR_auth-rbac-foundation.md`).

Physical naming convention: `snake_case` , per `ai/skills/database-design/SKILL.md` ("until [a convention is] fixed, use snake_case physical names consistently"). ASP.NET Core Identity's default PascalCase table names (`AspNetUsers` , etc.) are remapped to `snake_case` via EF Core Fluent API `ToTable(...)` calls in `AppDbContext` , and `EFCore.NamingConventions` (proposed in `work-items/WI-001/plan.md` , Auth foundation design) snake_cases column names consistently. Keys are `uuid` defaulting to `gen_random_uuid()` (the claim tables use Identity's `integer` identity keys), timestamps are `timestampz` , and constraint and index names follow EF Core's `pk_ / fk_ / ix_` style. 001_DB–003_DB follow this convention.

Document control (改版履歴)

Field	Value
Document ID	000_DB
System name	ProductionManagementAI
Work item	WI-001
Database	<code>production_management_ai</code> , schema <code>public</code>
DBMS	PostgreSQL 17
Character set	UTF-8
Created by	Claude (for ThanhTN)
Created date	2026-09-16
Last updated by	Claude (for ThanhTN)
Last updated date	2026-09-28

No	Target	Work item	Change	Date	Author
1	All	WI-001	Initial creation (as DB-001)	2026-09-16	Claude (for ThanhTN)
2	All	—	Template references updated	2026-09-17	ThanhTN
3	Open decisions	WI-002	CSRF decision recorded as decided (WI-002 DEC-020)	2026-09-18	Claude (for ThanhTN)

No	Target	Work item	Change	Date	Author
4	All	WI-005	Renamed to 000_DB (RFC 0010); English and Japanese PDFs added (RFC 0008)	2026-09-23	Claude (for ThanhTN)
5	All	—	Restructured to the table-definition template (document control, table list with lifecycle, per-table column definitions, index and trigger lists, database role section). Corrected against the <code>InitialIdentitySchema</code> migration: <code>users</code> column order, <code>phone_number</code> type (<code>text</code> , not <code>varchar(32)</code>), column defaults (the booleans and <code>access_failed_count</code> have no database default; the application sets them), FK names, the four FK indexes, and the uniqueness constraints (unique indexes, not separate <code>UNIQUE</code> constraints). The four Identity default tables are itemized; <code>user_claims</code> and <code>role_claims</code> recorded as read at sign-in	2026-09-28	Claude (for ThanhTN)

Table list (テーブル一覧)

No	Schema	Physical name	Logical name	Overview	Indexes beyond PK	Triggers	Created	Schema changed	Dropped	Status
1	public	users	Users	Application user accounts (Identity <code>IdentityUser<Guid></code> + app-specific fields)	yes	no	2026-09-16	—	—	in use
2	public	roles	Roles	RBAC roles (Identity <code>IdentityRole<Guid></code>); seeded <code>Admin</code> , <code>Operator</code>	yes	no	2026-09-16	—	—	in use
3	public	user_roles	User roles	Many-to-many join between users and roles	yes	no	2026-09-16	—	—	in use
4	public	user_claims	User claims	Identity default — per-user claims	yes	no	2026-09-16	—	—	in use — read at sign-in; empty, nothing writes it
5	public	role_claims	Role claims	Identity default — per-role claims	yes	no	2026-09-16	—	—	in use — read at sign-in; empty, nothing writes it
6	public	user_logins	User logins	Identity default — external login providers	yes	no	2026-09-16	—	—	unused (no external login in scope)
7	public	user_tokens	User tokens	Identity default — per-user tokens, e.g. password reset	no	no	2026-09-16	—	—	unused (kept for extensibility)

"Created" is the date of the migration that created the table (`InitialIdentitySchema` , 2026-09-16).

ER diagram and relationships

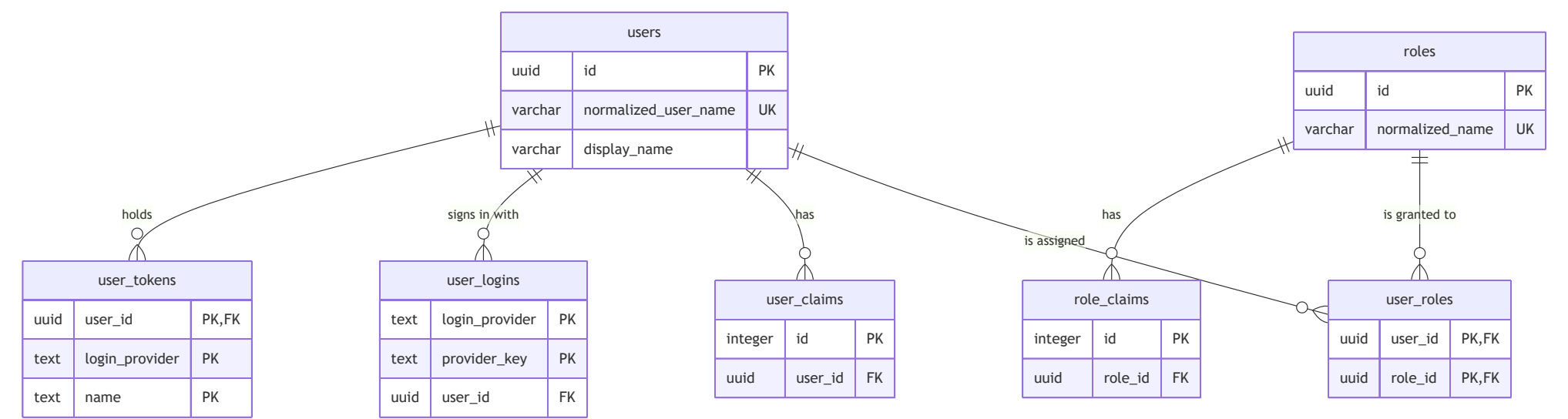


Table	Related table	Relationship (1:1 / 1:N / N:N)	FK column
users	roles	N:N (via user_roles)	user_roles.user_id / user_roles.role_id
users	user_claims	1:N (a user may have none)	user_claims.user_id
users	user_logins	1:N (a user may have none)	user_logins.user_id
users	user_tokens	1:N (a user may have none)	user_tokens.user_id
roles	role_claims	1:N (a role may have none)	role_claims.role_id

Table definitions (テーブル定義)

Validation for every Identity-managed column is ASP.NET Core Identity's own (`userManager` / `roleManager` and their options); no screen writes these tables. Where a column has no database default, the application supplies the value on insert (Identity's entity defaults or `AppUser` 's initializers).

Users (users)

Field	Value
Logical name	Users

Field	Value
Physical name	users
Schema	public
Overview	One row per application user account. Written only through ASP.NET Core Identity (UserManager): the development seed admin by IdentitySeeder , and login bookkeeping (lockout, failed count, security stamp). Columns follow IdentityUser<Guid> plus the app-specific display_name , is_active , created_at_utc (AppUser)
Estimated volume	small, bounded — plant staff accounts, tens of rows
Retention / deletion	kept; no deletion feature exists. Deleting a user would cascade to its role assignments, claims, logins and tokens

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	Id	id	uuid	—	yes	gen_random_uuid()	○			—	○	Identity IdentityUser<Guid> key	—	Returned by GET /api/auth/me
2	DisplayName	display_name	varchar	200	yes	none				Application sets it on create	○	Name shown in the UI header	—	App-specific addition
3	IsActive	is_active	boolean	—	yes	none (application sets true)				—		Account-enabled flag	true : active; false : disabled	App-specific addition; reserved for a future account-disable use case, not yet consumed by any requirement
4	CreatedAtUtc	created_at_utc	timestampz	—	yes	now()				—	○	When the account was created (UTC)	—	App-specific addition
5	UserName	user_name	varchar	256	no	none				Identity	○	Login name	—	POST /api/auth/login → userName
6	NormalizedUserName	normalized_user_name	varchar	256	no	none		○		Identity	○	Upper-cased user name for case-insensitive lookup	—	Unique via ix_users_normalized_user_name
7	Email	email	varchar	256	no	none				Identity		Email address	—	Identity default; no email-based feature yet
8	NormalizedEmail	normalized_email	varchar	256	no	none				Identity		Upper-cased email for lookup	—	Not unique (RequireUniqueEmail is off)
9	EmailConfirmed	email_confirmed	boolean	—	yes	none (application sets false)				Identity		Email confirmation flag	—	No email-confirmation flow in scope
10	PasswordHash	password_hash	text	—	no	none				Identity	○	Identity-managed password hash	—	Never read or written outside Identity APIs

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
11	SecurityStamp	security_stamp	text	—	no	none				Identity	○	Changes when credentials change, invalidating existing sessions	—	Identity default
12	ConcurrencyStamp	concurrency_stamp	text	—	no	none				Identity	○	Optimistic-concurrency token for Identity updates	—	Identity default
13	PhoneNumber	phone_number	text	—	no	none				Identity		Phone number	—	Identity default; unused
14	PhoneNumberConfirmed	phone_number_confirmed	boolean	—	yes	none (application sets false)				Identity		Phone confirmation flag	—	Identity default; unused
15	TwoFactorEnabled	two_factor_enabled	boolean	—	yes	none (application sets false)				Identity		2FA flag	—	No 2FA flow in scope
16	LockoutEnd	lockout_end	timestampz	—	no	none				Identity	○	End of the current lockout, if any	—	Supports the STRIDE lockout mitigation in 0002_ADR
17	LockoutEnabled	lockout_enabled	boolean	—	yes	none (Identity sets true for new users)				Identity	○	Whether this user can be locked out	—	Identity default
18	AccessFailedCount	access_failed_count	integer	—	yes	none (application sets 0)				Identity	○	Consecutive failed sign-ins	—	Reset on successful sign-in

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** one development admin user (admin , display name Seed Admin), created at startup by IdentitySeeder when SEED_ADMIN_PASSWORD is set, not by a migration (0002_ADR).

Roles (roles)

Field	Value
Logical name	Roles
Physical name	roles
Schema	public
Overview	RBAC roles (Identity IdentityRole<Guid>). Written only by IdentitySeeder through RoleManager
Estimated volume	small, bounded — the placeholder roles, 2 rows
Retention / deletion	kept; role deletion is not exposed by any feature

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	Id	id	uuid	—	yes	gen_random_uuid()	○			—	○	Identity IdentityRole<Guid> key	—	
2	Name	name	varchar	256	no	none				Identity	○	Role name	Admin ; Operator (placeholders, 0002_ADR)	Returned in GET /api/auth/me → roles
3	NormalizedName	normalized_name	varchar	256	no	none		○		Identity	○	Upper-cased role name for lookup	—	Unique via ix_roles_normalized_name
4	ConcurrencyStamp	concurrency_stamp	text	—	no	none				Identity	○	Optimistic-concurrency token	—	Identity default

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** Admin and Operator , created at startup by IdentitySeeder if missing, not by a migration.

User roles (user_roles)

Field	Value
Logical name	User roles
Physical name	user_roles
Schema	public
Overview	One row per role assigned to a user (Identity IdentityUserRole<Guid>)
Estimated volume	small, bounded — at most users × roles
Retention / deletion	physical delete when a role is removed from a user; cascades from users and roles

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	UserId	user_id	uuid	—	yes	none	○		users.id	—	○	Assigned user	—	Composite PK with role_id
2	RoleId	role_id	uuid	—	yes	none	○		roles.id	—	○	Assigned role	—	Composite PK with user_id

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** the seed admin's Admin assignment, created by IdentitySeeder .

User claims (user_claims)

Field	Value
Logical name	User claims
Physical name	user_claims
Schema	public
Overview	Identity default table for per-user claims. AddIdentity<AppUser, AppRole> registers Identity's UserClaimsPrincipalFactory , which reads this table for the signed-in user at every sign-in; no feature writes it, so it stays empty. Kept at near-zero cost so that claims, if needed later, need no schema change (work-items/WI-001/plan.md , Auth foundation design)
Estimated volume	empty
Retention / deletion	physical delete by Identity; cascades from users

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	Id	id	integer	—	yes	identity (GENERATED BY DEFAULT AS IDENTITY)	○			—		Surrogate key	—	Sequence user_claims_id_seq
2	UserId	user_id	uuid	—	yes	none			users.id	—	○	Owning user	—	Sign-in reads claims by user
3	ClaimType	claim_type	text	—	no	none				Identity	○	Claim type URI	—	
4	ClaimValue	claim_value	text	—	no	none				Identity	○	Claim value	—	

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** None.

Role claims (role_claims)

Field	Value
Logical name	Role claims
Physical name	role_claims
Schema	public
Overview	Identity default table for per-role claims. Read at every sign-in for each of the user's roles (the same principal factory as user_claims); no feature writes it, so it stays empty
Estimated volume	empty
Retention / deletion	physical delete by Identity; cascades from roles

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	Id	id	integer	—	yes	identity (GENERATED BY DEFAULT AS IDENTITY)	○			—		Surrogate key	—	Sequence role_claims_id_seq
2	RoleId	role_id	uuid	—	yes	none			roles.id	—	○	Owning role	—	Sign-in reads claims by role
3	ClaimType	claim_type	text	—	no	none				Identity	○	Claim type URI	—	
4	ClaimValue	claim_value	text	—	no	none				Identity	○	Claim value	—	

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** None.

User logins (user_logins)

Field	Value
Logical name	User logins
Physical name	user_logins
Schema	public
Overview	Identity default table linking a user to an external login provider. Unused: there is no external login in scope
Estimated volume	empty
Retention / deletion	physical delete by Identity; cascades from users

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	LoginProvider	login_provider	text	—	yes	none	○			Identity		External provider name	—	Composite PK with provider_key
2	ProviderKey	provider_key	text	—	yes	none	○			Identity		The user's key at that provider	—	Composite PK with login_provider
3	ProviderDisplayName	provider_display_name	text	—	no	none				Identity		Provider name for display	—	
4	UserId	user_id	uuid	—	yes	none			users.id	—		Owning user	—	

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** None.

User tokens (user_tokens)

Field	Value
Logical name	User tokens
Physical name	user_tokens
Schema	public
Overview	Identity default table for per-user tokens (e.g. password reset, authenticator keys). Unused by any feature yet
Estimated volume	empty
Retention / deletion	physical delete by Identity; cascades from users

No	Item name	Physical name	Data type	Length	NOT NULL	Default	PK	UK	FK	Validation	Used	Description	Code values	Notes
1	UserId	user_id	uuid	—	yes	none	○		users.id	—		Owning user	—	Composite PK with login_provider , name
2	LoginProvider	login_provider	text	—	yes	none	○			Identity		Token issuer	—	
3	Name	name	text	—	yes	none	○			Identity		Token name	—	
4	Value	value	text	—	no	none				Identity		Token value	—	

- **Usage patterns (パターン):** Not applicable — one kind of record.
- **CSV import/export (DL対象 / UL更新):** Not applicable — no CSV import/export.
- **Seed data:** None.

Index definitions (インデックス一覧)

All indexes were created by InitialIdentitySchema (2026-09-16) on new, empty tables, so none needed CONCURRENTLY .

No	Table	Index name	Method	Unique	Column(s) and sort order	Partial predicate / INCLUDE	Built concurrently	Created	Dropped	Rationale (access pattern)
1	users	pk_users	btree	yes	id ASC	—	no — new table	2026-09-16	—	Primary key; load a user by id (cookie principal)
2	users	ix_users_normalized_user_name	btree	yes	normalized_user_name ASC	—	no — new table	2026-09-16	—	Login looks up by normalized user name; Identity requires uniqueness
3	users	ix_users_normalized_email	btree	no	normalized_email ASC	—	no — new table	2026-09-16	—	Identity's email lookup; not unique because RequireUniqueEmail is off (the brief doesn't require email-based login)

No	Table	Index name	Method	Unique	Column(s) and sort order	Partial predicate / INCLUDE	Built concurrently	Created	Dropped	Rationale (access pattern)
4	roles	pk_roles	btree	yes	id ASC	—	no — new table	2026-09-16	—	Primary key
5	roles	ix_roles_normalized_name	btree	yes	normalized_name ASC	—	no — new table	2026-09-16	—	Role lookup by name; keeps seeding idempotent
6	user_roles	pk_user_roles	btree	yes	user_id ASC, role_id ASC	—	no — new table	2026-09-16	—	"What roles does this user have", at sign-in and on GET /api/auth/me
7	user_roles	ix_user_roles_role_id	btree	no	role_id ASC	—	no — new table	2026-09-16	—	EF Core FK convention; serves the cascade from roles and "users in role"
8	user_claims	pk_user_claims	btree	yes	id ASC	—	no — new table	2026-09-16	—	Primary key
9	user_claims	ix_user_claims_user_id	btree	no	user_id ASC	—	no — new table	2026-09-16	—	EF Core FK convention; sign-in loads the user's claims
10	role_claims	pk_role_claims	btree	yes	id ASC	—	no — new table	2026-09-16	—	Primary key
11	role_claims	ix_role_claims_role_id	btree	no	role_id ASC	—	no — new table	2026-09-16	—	EF Core FK convention; sign-in loads each role's claims
12	user_logins	pk_user_logins	btree	yes	login_provider ASC, provider_key ASC	—	no — new table	2026-09-16	—	Find a user by external login
13	user_logins	ix_user_logins_user_id	btree	no	user_id ASC	—	no — new table	2026-09-16	—	EF Core FK convention; logins by user
14	user_tokens	pk_user_tokens	btree	yes	user_id ASC, login_provider ASC, name ASC	—	no — new table	2026-09-16	—	Token lookup by user, provider and name; its leading column also serves the FK

Trigger definitions (トリガー一覧)

None — no triggers.

Constraints

Constraint	Type (PK/FK/UNIQUE/CHECK/EXCLUDE)	Table.column(s)	References	Rule
pk_users	PK	users.id	—	—

Constraint	Type (PK/FK/UNIQUE/CHECK/EXCLUDE)	Table.column(s)	References	Rule
pk_roles	PK	roles.id	—	—
pk_user_roles	PK	user_roles.(user_id, role_id)	—	—
pk_user_claims	PK	user_claims.id	—	—
pk_role_claims	PK	role_claims.id	—	—
pk_user_logins	PK	user_logins.(login_provider, provider_key)	—	—
pk_user_tokens	PK	user_tokens.(user_id, login_provider, name)	—	—
fk_user_roles_users_user_id	FK	user_roles.user_id	users.id	ON DELETE CASCADE — removing a user removes their role assignments
fk_user_roles_roles_role_id	FK	user_roles.role_id	roles.id	ON DELETE CASCADE — removing a role removes its assignments (role deletion itself is not exposed by any feature yet)
fk_user_claims_users_user_id	FK	user_claims.user_id	users.id	ON DELETE CASCADE
fk_role_claims_roles_role_id	FK	role_claims.role_id	roles.id	ON DELETE CASCADE
fk_user_logins_users_user_id	FK	user_logins.user_id	users.id	ON DELETE CASCADE
fk_user_tokens_users_user_id	FK	user_tokens.user_id	users.id	ON DELETE CASCADE

Uniqueness of `users.normalized_user_name` and `roles.normalized_name` is enforced by the unique indexes `ix_users_normalized_user_name` and `ix_roles_normalized_name` (index list Nos. 2 and 5); there is no separate `UNIQUE` constraint. Both columns are nullable and PostgreSQL treats `NULL` s as distinct in a unique index, but Identity always sets them from the user or role name, so this never arises.

Database role and privileges

When WI-001 created these tables, the backend used the owner login for everything. Since WI-002 (001_DB, DEC-016), the backend runs as the restricted `pmai_app` login. The `AddProductionOrders` migration grants that login the following on these tables:

Role	Used by	Privileges on these tables	Reason
<code>pmai_app</code> (runtime login)	The API at runtime, including <code>IdentitySeeder</code>	SELECT, INSERT, UPDATE on all seven tables; DELETE on <code>user_roles</code> , <code>user_claims</code> , <code>user_logins</code> , <code>user_tokens</code> ; USAGE, SELECT on sequences <code>user_claims_id_seq</code> , <code>role_claims_id_seq</code>	What <code>UserManager</code> , <code>RoleManager</code> and <code>SignInManager</code> issue: sign-in bookkeeping updates <code>users</code> , seeding inserts users, roles and assignments, and Identity removes rows from the four child tables. No DELETE on <code>users</code> / <code>roles</code> , no DDL, no TRUNCATE
Owner (<code>POSTGRES_USER</code>)	Migrations only (<code>dotnet ef database update</code>)	owner — DDL	Creates and alters the schema

API / DD mapping

Table.column	DD field	API field
users.id	not yet written (WI-001 has no DD; auth endpoints are specified directly in 0002_ADR)	GET /api/auth/me → id
users.user_name	—	GET /api/auth/me → userName ; POST /api/auth/login request → userName
users.display_name	—	GET /api/auth/me → displayName
roles.name (via user_roles)	—	GET /api/auth/me → roles: string[]

Migration impact and recovery limits

Migration	Type	Tables touched	Live table?	Sequence
InitialIdentitySchema (2026-09-16)	additive	all seven tables above (created)	no — the project's first migration	single step

- **Data recovery limit:** not applicable at creation — there were no pre-existing rows. Reverting it today would delete every user account and role assignment; they are recreated only for the seed admin.
- **Rollback plan:** dotnet ef database update 0 drops these tables. Since WI-002, later migrations grant on these tables and depend on them, so this is only possible after reverting every later migration first.

Open decisions

Decision	Options	Recommendation	Status
Whether an additional CSRF token (beyond SameSite cookie attribute) is needed for state-changing endpoints	SameSite=Lax only vs. SameSite=Lax + anti-forgery token	SameSite=Lax only — no additional token	decided — work-items/WI-002/decisions.md DEC-020 (2026-09-18)
Whether email_confirmed / two_factor_enabled / phone_number* columns are ever exercised by a future requirement	keep unused vs. trim from the model	Keep (near-zero cost, avoids a later breaking migration) — already decided, see the users table notes	decided — see work-items/WI-001/decisions.md (auth foundation design)